

## 1.1 NLP in the Real World

What NLP is, the core tasks, and why human language is a hard input.

### 1. What NLP is

Natural language processing (NLP) is how we get a computer to work with **human language**: read it, write it, search it, translate it, or answer a question about it.

It sits between linguistics, computer science, and artificial intelligence. The input is messy text (or speech turned into text). The output is a label, a span, a ranking, another sentence, or a number.

This course is **applied**. The goal is to build systems that do a job, not to survey every theory of language.

The main text for the ideas this week is *Practical Natural Language Processing* (Vajjala, Majumder, Gupta, Surana).

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### 2. Tasks you will see all semester

A small set of jobs shows up in almost every project:

| Task                   | What you ask the system to do                               |
|------------------------|-------------------------------------------------------------|
| Language modeling      | Predict the next word from the words so far                 |
| Text classification    | Put a document into a known bucket (spam, topic, sentiment) |
| Information extraction | Pull out people, dates, events, or other fields             |
| Information retrieval  | Find documents that match a query                           |
| Conversational agent   | Keep a dialogue going in ordinary language                  |
| Text summarization     | Shorten a document and keep the point                       |
| Question answering     | Answer a question written in English                        |
| Machine translation    | Rewrite the text in another language                        |
| Topic modeling         | Find themes in a large collection of documents              |

Later weeks pick these up one at a time. Week 1 is the map.



Text  
Classification



Information  
Extraction



Conversational  
Agent



Information  
Retrieval



Question  
Answering Systems



Spam  
Classification



Calendar Event  
Extraction



Personal  
Assistants



Search  
Engines

**JEOPARDY!**

Jeopardy!



Social Media  
Analysis



Retail Catalog  
Extraction



Health Records  
Analysis



Financial  
Analysis



Legal Entity  
Extraction

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### 3. Why the job is hard

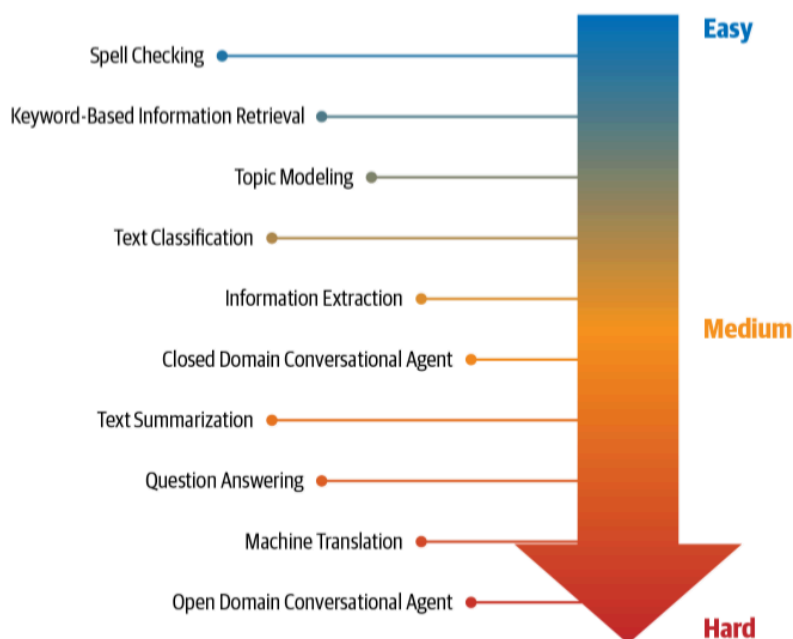
Human language is not a clean table.

**Ambiguity.** The same string can mean more than one thing. *I saw her duck* is a bird or a movement. A model has to use context.

**Common knowledge.** Speakers leave facts unsaid. "The restaurant was packed, so we left" assumes you know what a packed restaurant is like. The text does not spell that out.

**Creativity.** Dialects, slang, sarcasm, and new coinages break rules you wrote last month.

**Diversity across languages.** There is no one-to-one map between vocabularies. A pipeline that works for English does not port by renaming a folder.



The man couldn't lift his son because he was so **weak**. — Who was weak?

The man couldn't lift his son because he was so **heavy**. — Who was heavy?

Mary and Sue are **sisters**.  
 Mary and Sue are **mothers**. — How are Mary and Sue related?

Joan made sure to thank Susan for all the help she had **received**. — Who had received help?

Joan made sure to thank Susan for all the help she had **given**. — Who had given help?

John **promised** Bill to leave, so an hour later he left.  
 John **ordered** Bill to leave, so an hour later he left. — Who left an hour later?

## 4. What this week is for

By the end of Week 1 you should be able to:

- name the core NLP tasks in the table above
- say why a rule-only system struggles
- write a short Python script that uses lists, dictionaries, functions, and files

Note **1.2** is how machine learning and deep learning sit next to those tasks. Note **1.3** is the Python you need for Lab 1.

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## **5. Practice**

1. Pick a product you use (search, mail, maps, a chatbot). Name two NLP tasks from the table that it is doing.
2. Write one English sentence that is ambiguous. Say the two readings.
3. Why would a perfectly good English spam filter do poorly on another language even if you translate the training labels?